# OTA updates




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Corrections, suggestions, contributions and translations are welcome!



# Runtime Package Management



# Runtime Package Management

- BitBake always builds packages selected in IMAGE_INSTALL.
- The binary packages are used to generate the root filesystem.
- It is also possible to update the system at runtime using these packages, for many use cases:
  - In-field security updates.
  - System updates over the wire.
  - System, packages or configuration customization at runtime.
  - Remote debugging.
- Using the Runtime Package Management is an optional feature.
- Can be useful for local development, but rarely used in production.
- See the extra slides for details on how to set it up.





# A/B updates



# A/B updates: principle

- Mechanism for Over-The-Air (OTA) *image* updates
- Key idea: avoid one update bricking the device
  - Even if tested, one update might prevent the system from booting
  - Enable the device to recover from a bad update automatically
- The key idea is to maintain at least two copies, in separate partitions (slots) of your storage device
  - minimum is 2 rootFS
  - usually also includes the kernel
  - can involve the bootloader, depending on the ROM code



# A/B updates: Yocto support

- Three main open-source solutions:
  - SWUpdate:
    - DIY approach, very flexible
    - Yocto layer: meta-swupdate
  - Mender:
    - Fully-integrated solution, requires `systemd`
    - Yocto layer: meta-mender
  - RAUC:
    - Good middleground
    - Yocto layer: meta-rauc
- See our Security training for in-depth details on A/B updates.